

Use the following context to answer questions 1-5.

The average mass of a baseball is $145\frac{1}{2}$ grams (g) with a tolerance of $3\frac{1}{2}$ g. If x is the mass of the baseball and V is the variation from $145\frac{1}{2}$ g., the function $V(x) = \left|x - 145\frac{1}{2}\right|$ can be used to find the amount of variation.

1. Determine the maximum and minimum x -values.

minimum = 142 grams

maximum = 149 grams

2. Complete the table for $V(x) = \left|x - 145\frac{1}{2}\right|$. Use the minimum x -value in the first column, the maximum x -value in the last column, and add $1\frac{3}{4}$ to each x -value in between.

x	142	$143\frac{3}{4}$	$145\frac{1}{2}$	$147\frac{1}{4}$	149
y	$3\frac{1}{2}$	$1\frac{3}{4}$	0	$1\frac{3}{4}$	$3\frac{1}{2}$

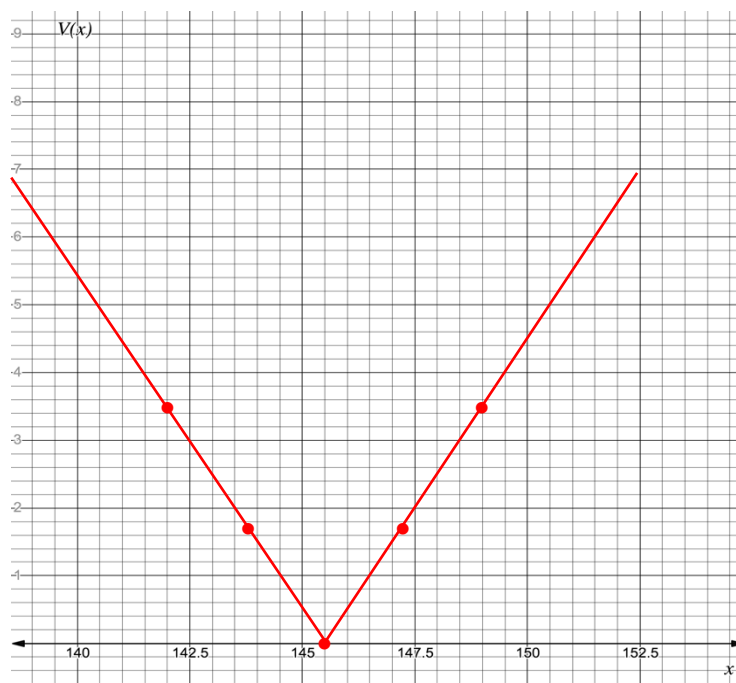
3. Sketch the graph of the absolute value function

4. Interpret the vertex of $V(x) = \left|x - 145\frac{1}{2}\right|$.

The average mass of a baseball.

5. Interpret the axis of symmetry of $V(x) = \left|x - 145\frac{1}{2}\right|$.

The line that determines how far away from the average mass of a baseball, in grams, in either direction the mass varies.



Use the following context to answer questions 6-10.

The clearance below a bridge is 185.5 feet (ft) above sea level. According to the distribution of tidal phases, the water under the bridge can fluctuate up to 3 feet above or below the normal depth. If x is the amount of clearance the bridge has above the water, in ft, and W is the variation in water depth away from sea level, in ft, the equation to find the amount of variation is $W(x) = |x - 185.5|$.

6. Determine the maximum and minimum clearance between the bridge and water.

minimum = 182.5 feet maximum = 188.5 feet

7. Complete the table for $W(x) = |x - 185.5|$.

x	182.5	183.5	184.5	185.5	186.5	187.5	188.5
y	3	2	1	0	1	2	3

8. Sketch the graph of the absolute value function.

9. Interpret the vertex of $W(x) = |x - 185.5|$.

No fluctuation above or below the normal depth.

10. Interpret the axis of symmetry of $W(x) = |x - 185.5|$.

The line that determines how far away from the normal depth, in feet, the water fluctuates.

